SIMATS ENGINEERING
Department of Computer Science and Engineering (Cyber Security)
ASSESSMENT TOOL2 - CONCEPT MAPPING

Course Code:	CSA1205	**Course Title:**	Computer Architecture for Multicore Systems
CO Assessed:	CO1	**Assessment:**	ASSESSMENT TOOL 2- CONCEPT MAPPING
Weightage:	35%	**Total Marks:**	25
CO1 Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)		
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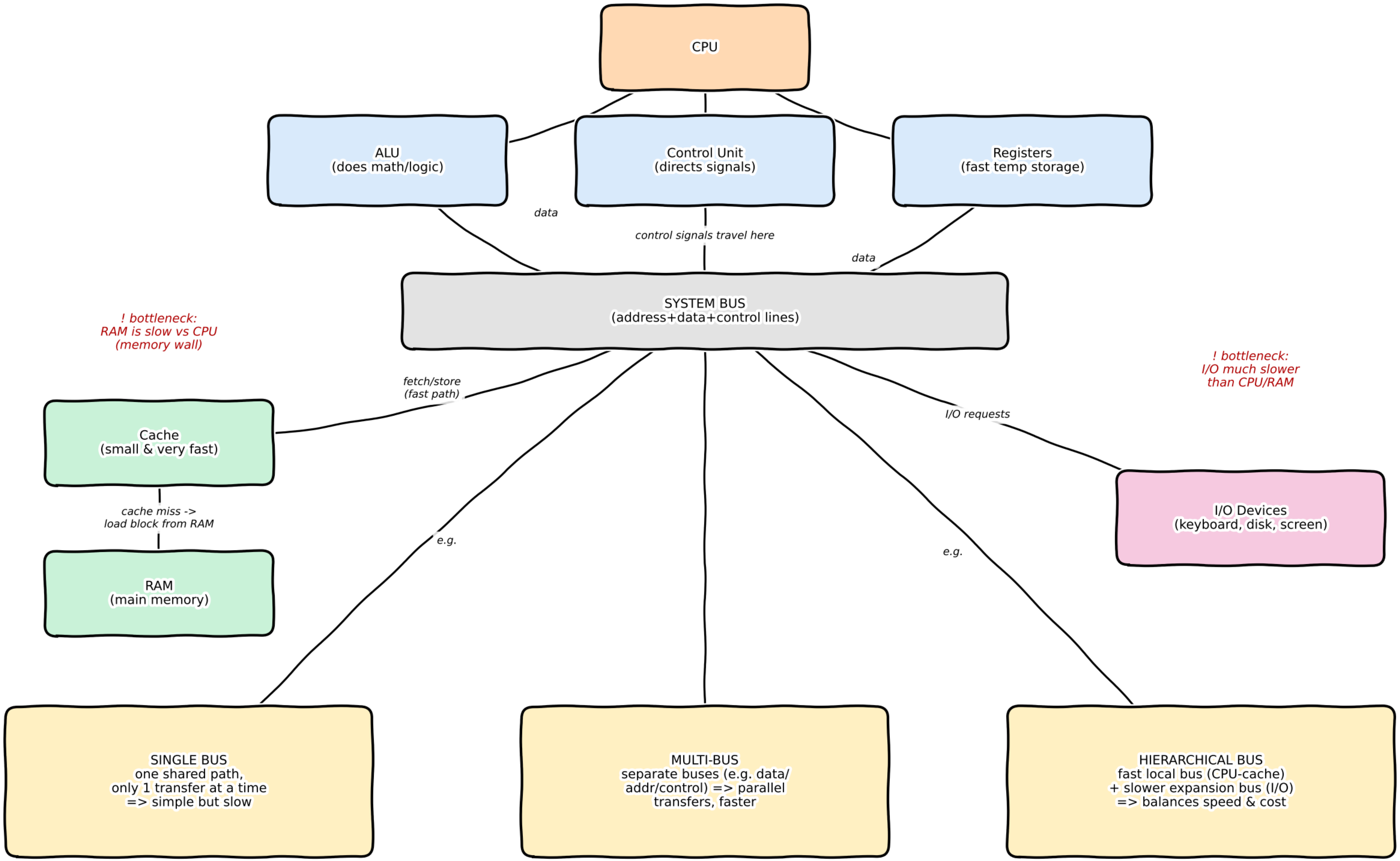
Department:	CSE Cyber Security	**Date:**	13-07-2026

Instruction to Students

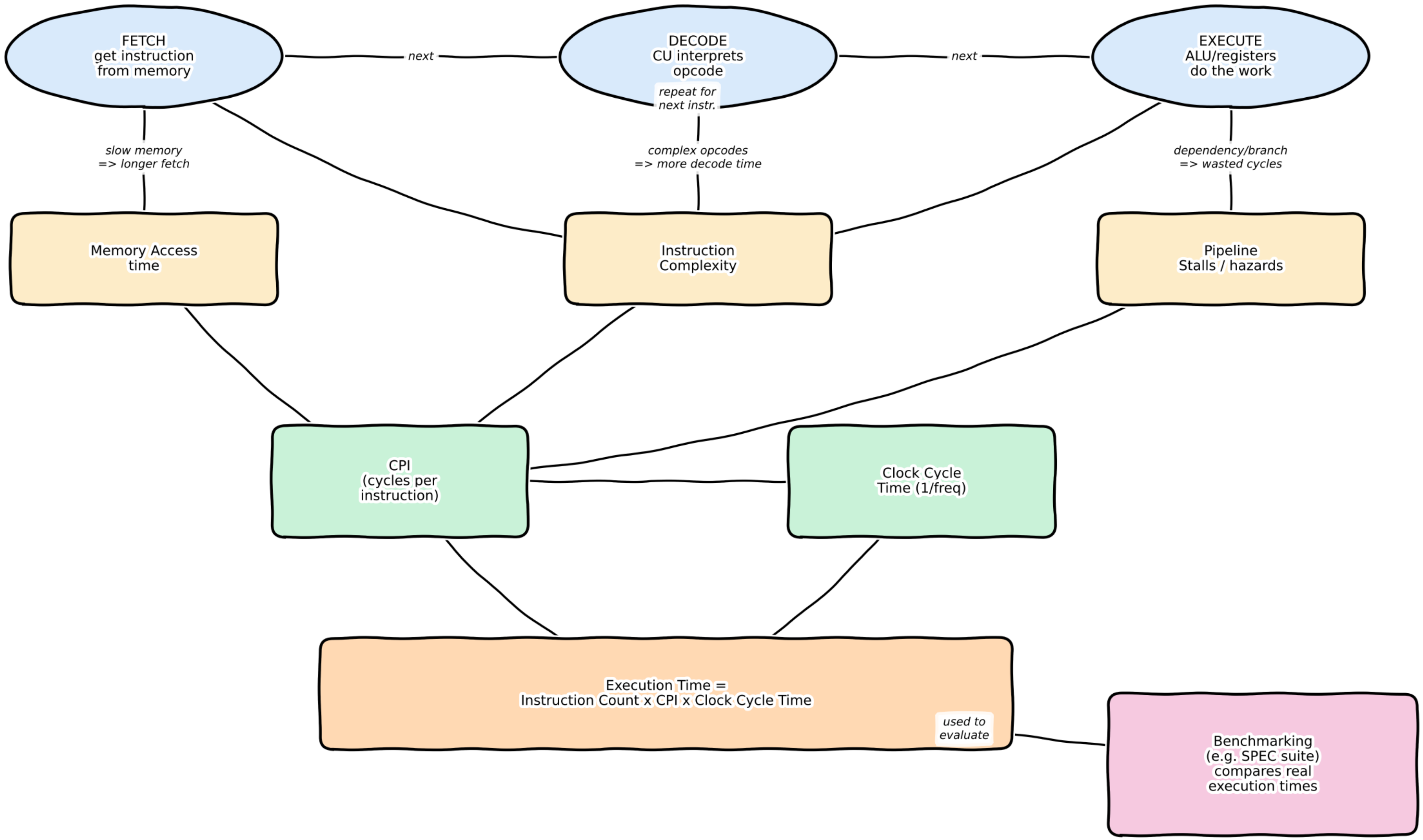
1. Answer two questions as per the Instruction.
2. Each question carries 12.5 mark.
3. Only the appropriate Tools can be used
4. Time allotted for the exam is 60 min

Q1: CPU - Memory - I/O and Bus Structures



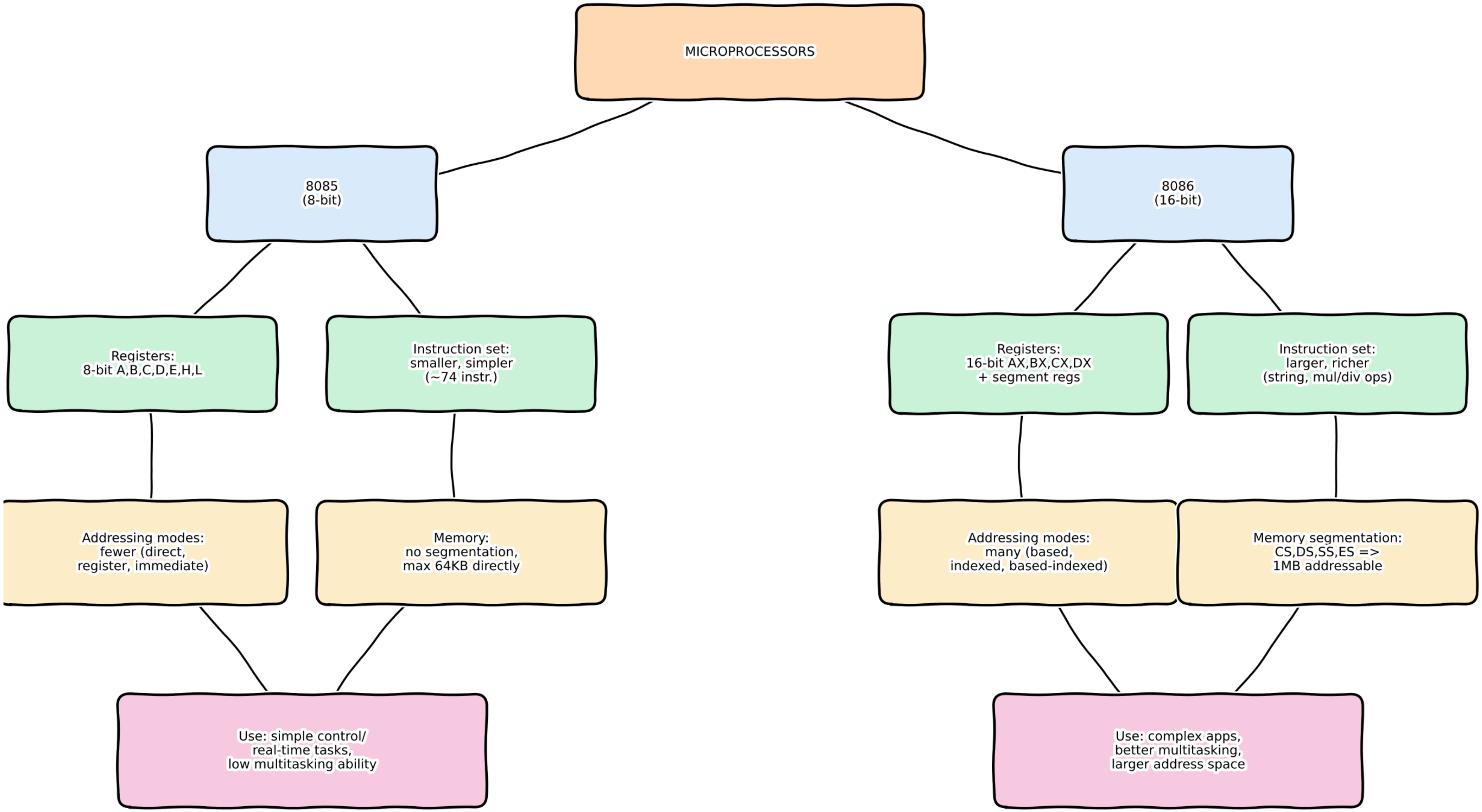
Note: wider / more buses = less contention = better performance, but higher cost & complexity

Q2: Instruction Cycle and CPU Performance



Lower execution time = better performance | fewer stalls & simpler instr. => lower effective CPI

Q3: 8085 vs 8086 Architecture Comparison



Segmentation + richer addressing + wider registers => 8086 handles bigger, multitasked, real-world programs better

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

Criteria	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)
Q1					
Q2					

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand